A Sparse Sharing Multitask Framework for Building Footprint Extraction From Remote Sensing Imagery Following the Dual Lottery Ticket Hypothesis

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***Abstract*— Building footprint extraction from high-resolution remote sensing imagery is significant for urban planning, change detection, disaster management, and other applications. Recently, researchers have found that the edge features of buildings are crucial in extracting building footprints, and multitask deep learning is used to share edge feature information. However, these multitask deep learning frameworks adopt a hard sharing approach, which cannot avoid the adverse effects caused by the differences between different tasks, resulting in the problem of blurred edges and building boundaries. To address this issue, this article proposes a dual lottery ticket hypothesis (DLTH) and sparse sharing-based multitask deep learning framework, dual sparse sharing architecture (DSSA), to transmit the edge information in the edge detection to the building footprint extraction by sharing partial parameters. First, the subnetworks of building footprint extraction and edge detection are constructed according to the sparse rate and parameter sharing rate to control the dependencies between the subnetworks. Second, given the difference in the importance of the two tasks, a cosine unequal-scaled alternating training strategy is proposed to strengthen and weaken the transmission of edge information periodically. Third, following the DLTH, the loss function with $L3/2$ regularization constraint is used to promote the information transmission and parameter conversion of the subnetwork by using the global information. Finally, aiming at the edge of building footprint extraction results, a pixel-based evaluation index, edge extraction accuracy ($EEA^{(n)}$), is designed by morphological erosion to better evaluate the integrity of the edge of building footprint extraction results. The experiments conducted on a self-annotated dataset and two public datasets (i.e., WHU Aerial Imagery dataset and Massachusetts Building dataset) show that DSSA can achieve better edge effects than the baseline and show excellent generalization ability.**

***Index Terms*— Building footprint extraction, dual lottery ticket hypothesis (DLTH), multitask, sparse sharing.**

I. Introduction

REMOTE sensing technology plays a vital role in gathering data about the Earth's surface and providing a better understanding of it. The resolution of available remote sensing imagery is increasing annually due to the rapid development of this technology. High-resolution remote sensing imagery is rich in information and assists in the detailed extraction of ground objects such as buildings, roads, and water bodies. The spatial distribution of buildings is crucial for constructing cities, and automatic building footprint extraction is necessary in various fields such as urban planning, change detection, map service, and disaster management [1], [2], [3], [4], [5]. Buildings, roads, and artificial surfaces [6] have similar spectral characteristics, while different roof building materials exhibit significantly different spectral characteristics. This leads to the problem of blurred edges and incomplete boundaries in building footprint extraction. Enhancing the edge information is essential for accurately identifying the parts of the edge of a building, which still remains a challenge.

With the rapid advancement of neural networks and deep learning techniques, methods for building footprint extraction have transitioned from traditional approaches to deep learning. Contrary to conventional methods that involve manual feature engineering, deep learning harnesses the power of extensive data-driven techniques, enabling the automatic acquisition of high-level and hierarchical features, thus demonstrating clear advantages in increasingly complex urban scenarios. Many experts and scholars have proposed a series of convolutional neural network (CNN)-based methodologies for building footprint extraction, essentially involving pixel-level classification of images to obtain binary representations encompassing buildings and nonbuildings. Typical architectures include an encoder–decoder architecture, where the encoder learns deep architectural features of buildings, and the decoder employs these features for pixel-level classification [7].

Many studies have focused on designing efficient encoder or decoder architectures to improve the accuracy of building

Manuscript received 22 April 2024; revised 15 May 2024 and 12 June 2024; accepted 20 June 2024. Date of publication 24 June 2024; date of current version 5 July 2024. This work was supported in part by Shandong Provincial Natural Science Foundation under Grant ZR2022YQ36, in part by the Program of China Scholarship Council under Grant 202209995003, in part by the Youth Innovation Team Project of Higher School in Shandong Province under Grant 2022KJ201, and in part by the National Natural Science Foundation of China under Grant 41871283. *(Corresponding authors: Junwu Xiang; Qi Wen.)*

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Digital Object Identifier 10.1109/TGRS.2024.3418369

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footprint t extraction [8], [9], [10], [11], [12], [13]. Many regard U-Net [14] as one of the most iconic and classical encoder–decoder footprint extraction networks. The encoder is used to downsample and extract feature information at different scales, while the decoder is responsible for upsampling and feature fusion. RU-Net [15] incorporates an improved residual network [16] into the U-Net architecture to reduce parameters and prevent network degradation. Employing the atrous spatial pyramid pooling (ASPP) model [17] as the intermediate unit connecting the encoder and decoder units facilitates extracting and connecting multiscale features and contextual information easy. The encoder sections of the deep cross-modal gated fusion network (CMGFNet) [18] are built on a residual network for both RGB and digital surface model (DSM) streams. The feature of the RGB encoder guides the learning of the DSM feature to obtain cross-modal feature fusion through the gated fusion module (GFM). Based on the U-Net framework, Yang et al. [19] quantitatively compared the different depths and architectures of classic backbones (including VGG13, VGG19 [20], ASPP-based VGG19, Res Net18, Res Net50 network [16], Dark Net19 [21], and Dark Net53 [22]). These methods significantly improve the accuracy of building footprint extraction from remote sensing imagery. However, the above techniques mainly target each pixel for classification and seldom consider using building edge information. Although there is an improvement in the overall accuracy, blurry edges, and incomplete boundaries persist in the building footprint extraction.

With the deepening of the research, researchers have discovered that because the pixels are independent of each other in the extraction of building footprints [8], this approach cannot solve the problem of missing small buildings and holes in large buildings. Buildings usually have some regular geometry or a combination of geometries. Therefore, it is essential to make full use of the contour information of the building. It is common practice to apply conditional random fields (CRFs) [23] to postprocess building footprint extraction results to improve the accuracy of boundary contour annotation further. This method is simple in design but has limited scope for improvement [24]. Using feature pairwise CRF (FPCRF) as a graph model can effectively generate clear boundaries and fine-grained building footprint extraction results, preserving the integrity of buildings and filtering some nonbuildings erroneously detected as buildings by CNN [25]. Employing a multiscale receptive field graph attention network (GAT) can effectively extract local-global adjacent node features and edge features [26]. Furthermore, GAT is capable of learning the importance of neighborhoods of different sizes, suppressing noise nodes for classification [27]. In addition, some researchers used a deep learning edge detection approach to prioritize boundary information and go beyond footprint extraction to obtain building contours directly. Wu et al. [28] proposed a BRNet architecture to predict segmentation and contours using local and global information to achieve better building segmentation and more accurate contour extraction. Structural similarity is used for edge information to evaluate predicted and actual boundary pixels and reduce spotted fragments [29]. Based on the architecture of bidirectional cascade network (BDCN) [30], a multiscale erosion network (ME-Net) [31] is embedded with erosion modules to erode the pixels on the outermost edge to detect crisp building edges. Jiwani et al. [32] used a feature-based approach by introducing a pyramid network (FPN) [33] module to achieve cross-scale feature extraction. They designed a particular weighted boundary map to penalize incorrect building boundary prediction, thus improving the DeepLabV3+ [34] model. DCEFE-Net [35] iteratively superimposes edge features through its edge-aware feature enhancement (EAFE) module and addresses scale variations among buildings by employing upsampling based on network depth. Furthermore, the connectivity between network modules is enhanced by incorporating densely connected convolutional layers. This design strengthens the interconnections between modules, thereby improving the robustness of the edge extraction network. However, due to the complex background in remote sensing imagery, artificial nonbuilding features share similar edge features with artificial buildings. Accurate identification of building edges remains a significant challenge.

Although there is a certain degree of nonlinearity and spatial interdependence between remote sensing data [36], the linear characteristics and spatial properties of remote sensing imagery are equally important. Building footprint extraction and building edge detection have a strong correlation. Building footprint extraction can provide building location information for edge detection to help it determine the edge of the building. Edge detection can provide building edge information for footprint extraction. Edge information helps it improve edge integrity. Multitask learning (MTL) aims to solve various tasks with branch feature-sharing architecture [37]. Bischke et al. [38] adopted a multitask deep learning framework to combine the learning of boundary and semantic information to improve the accuracy of the edges of footprint extraction results. Yin et al. [39] fully considered the existing remote sensing image-building extraction tasks and designed a multitask deep learning framework that integrates building footprint extraction, edge detection, and object detection, pointing out the direction for multitask building extraction. Wu et al. [40] proposed a new method for building vectorization, multitask edge detection (MTED), which designs an MTL strategy to integrate building footprint into the METD framework and uses spatial context to guide edge detection closely. However, these methods use parameter hard sharing to achieve multitask training without considering the negative impact of constraints between different tasks. As mentioned above, multitask deep learning methods have been widely used in building extraction tasks, but these methods use parameter hard sharing to achieve multitask training, which cannot avoid the negative impact of differences between different tasks. However, sparse sharing [41] solves the problems faced by hard sharing, soft sharing, and hierarchical sharing by partially sharing parameters and avoids negative transfer in MTL but lacks a sparse sharing multitask deep learning framework for building footprint extraction and edge detection.

Inspired by the above observations, this article proposes a multitask deep learning framework called dual sparse sharing architecture (DSSA), which includes two tasks: building

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footprint extraction and edge detection. The two subnetworks are controlled by sparse rate and parameter sharing rate to achieve constrained sparse sharing. In the training phase, the cosine unequal-scaled alternating training strategy is used to periodically strengthen and weaken the transfer of edge detection information to building footprint extraction, which can effectively avoid negative transfer. According to the dual lottery ticket hypothesis (DLTH), the loss function with $L3/2$ regularization constraint is used to make full use of the global information in the multitask network to transform the subnetwork into the optimal subnetwork. Aiming at the edge of the building footprint extraction result, an evaluation index edge extraction accuracy ($EEA^{(n)}$) is designed by morphological corrosion to evaluate the integrity of the edge of the building footprint extraction result.

The main contributions of our study are summarized as follows.

1) The sparse rate and parameter sharing rate are introduced to control the generation of building footprint extraction and edge detection subnetworks to control the dependence between subnetworks better.
2) A DLTH-based $L3/2$ regularization constraint loss function is proposed, which uses the global information in multitasks to promote subnetworks training.
3) A cosine unequal-scaled alternating training strategy is proposed to periodically strengthen and weaken the information transmission between tasks and avoid the negative impact of mutual containment between different tasks.
4) An evaluation index $EEA^{(n)}$ is designed by using morphological corrosion to evaluate the integrity of the edge of the building footprint extraction results.

The remainder of this article is arranged as follows. Section II reviews the related work. Section III introduces the details of DSSA. Section IV introduces the dataset and experimental details. In Section V, comparative experiments and ablation experiments are carried out to prove the effectiveness of the proposed method. Section VI is discussed. Finally, Section VII summarizes this article.

II. Related Works

This section illustrates the related deep learning-based building extraction, including two parts: single-task building footprint extraction and multitask building footprint extraction.

A. Single-Task Building Footprint Extraction

In order to solve the problems of blurry edges in buildings of complex shapes, omission of small buildings, and holes in large-scale buildings, a large number of experts have conducted related research on building footprint extraction. CBR-Net [42] uses a module that gradually refines building predictions in a coarse-to-fine manner and a novel boundary refinement (BR) module to enhance the ability to perceive and refine building edges. Inspired by HRNetv2 [43], MAP-Net [44] learns multiscale features retained through spatial localization through multiple parallel paths, where each stage is gradually generated to extract high-level semantic features as a fixed resolution. Then, the attention module adaptively squeezes the channel features extracted from each path to achieve optimized multiscale fusion, and the pyramid space pool module captures global dependencies to refine discontinuous building footprints. ASF-Net [45] uses a deepened spatial upsampling block to filter useful information and help establish boundaries. An adaptive information utilization block is proposed to expand the receptive field of the feature map and refine the incomplete building footprint. The ability of the network to extract small buildings and building edges is improved. FD-Net [46] utilizes two key types of building information in the input image, including semantic information related to building integrity and edge information to improve building boundaries, aiming to improve the existing encoder–decoder framework by decoupling image features into edge subspace and semantic subspace. Then, considering the heterogeneity between edge features and semantic features, the decoupled features are integrated through a supervised guided fusion process. In addition, a lightweight and effective global context attention module is introduced to capture contextual building information, thereby enhancing feature representation.

The above methods mainly enhance the ability of perception and building edges by designing different structures. However, building footprint extraction cannot take advantage of the rich edge information brought by the edge detection process, which limits its ability to capture the building contour information, resulting in unsatisfactory edge extraction results.

B. Multitask Building Footprint Extraction

Some experts and scholars have used a multitask architecture to construct tasks such as footprint extraction and edge detection into a multitask network to improve the performance of each task. Bischke et al. [38] used a multitask deep learning framework to combine the learning of boundaries and semantic information to improve the footprint extraction of architectural boundaries. Ekim and Sertel [47] introduced an edge output to the footprint extraction process. Yin et al. [39] designed a multitask deep learning framework incorporating building footprint extraction, edge detection, and object detection by fully considering existing remote sensing imagery building extraction tasks. The semantic feature-constrained change detection (SFCCD) network [48] simultaneously performs feature extraction, footprint extraction, and change detection, with change detection as the main task and footprint extraction as the auxiliary task. High-level features guide the training of low-level feature maps, and multiscale features are fused. Multiple convolutions provide different receptive fields. MTPA-Net [49] is a multitask deep learning framework with an attention module, which overcomes the semantic gap in building extraction by introducing the concept of "scene." The input image is classified into several different scene categories. Subsequently, the scene-specific decoder network predicts the classification map based on the scene prior information provided by the classification network, which can effectively improve the accuracy of building extraction. In the line segment collaborate segmentation (LCS) framework [50], footprint extraction provides location guidance

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for vector extraction. In contrast, vector extraction provides precise positioning for footprint extraction, giving full play to the respective advantages of the two tasks. ALNet [51] enhances weakly supervised semantic segmentation of buildings by incorporating a feature-to-image restoration branch as an auxiliary task and using a semantic-aware multilevel feature progressive aggregation module to refine feature fusion, thereby improving segmentation precision.

The above network proves that each task can thoroughly learn the characteristics of other tasks through appropriately designed network structures. MTL can extract buildings more accurately than other methods. However, the structures used are hard shared architecture [41], without considering the harmful effects of mutual restraint between different tasks, and the effect of improving the fuzzy and incomplete extraction of building boundaries is still not ideal.

III. Dual Sparse Sharing Architecture

Compared to the mainstream building extraction multitask architecture, DSSA is a sparse sharing architecture in multitask architecture; the parameters are partially shared.

For building extraction tasks T that include footprint extraction and edge detection, its base network is F. Then, the parameters of the subnetwork associated with task t become $M_t \odot \boldsymbol{\theta}$, where $\odot$ denotes element-wise multiplication. In this way, each subnetwork can be uniquely represented as $F_t = F(D; M_t \odot \boldsymbol{\theta})$, where M_t is a binary mask matrix and satisfies $M_t \in \{0, 1\}$, the parameter $\boldsymbol{\theta}$ denotes the base network parameters, and D denotes the dataset used by task T.

DSSA belongs to a more general multitask deep learning framework. In order to better explain and verify the effectiveness of DSSA, based on the above theory, we choose two building extraction tasks, footprint extraction and edge detection, for description. The subnetwork of footprint extraction F_{Seg} and the subnetwork of edge detection F_{Edge} are shown in the following equation:

$$\begin{cases} F_{\text{Seg}} = F(D; M_{\text{Seg}} \odot \boldsymbol{\theta}) \\ F_{\text{Edge}} = F(D; M_{\text{Edge}} \odot \boldsymbol{\theta}) \end{cases} \tag{1}$$

where $F(\cdot)$ is the base network and M_{Seg} and M_{Edge} are the binary mask for footprint extraction and edge detection, respectively.

DSSA consists of three parts (as shown in Fig. 1), that is: 1) generating subnetworks for each task; 2) cosine unequal-scaled alternating training strategy; and 3) random sparse transformation (RST). Initially, masks are generated for footprint extraction and edge detection based on the base network. These masks are then element-wise multiplied with the base network to obtain the subnetworks. Finally, during the training subnetworks, RST and the cosine unequal-scaled alternating training strategy are utilized to complete the training process.

A. Generating Subnetworks for Each Task

Within the framework of sparse sharing architecture, the foundational network is engineered to be overparameterized, ensuring an ample hypothesis space to concurrently accommodate multiple task solutions. This design philosophy aligns with the lottery ticket hypothesis, which asserts that contemporary mainstream baseline networks are typically overparameterized. Pursuant to the DLTH, while randomly selected subnetworks have the potential to transform into winning "tickets," the direct random generation of subnetworks for multifarious tasks may prove suboptimal. Specifically, when a reduced sparse rate is employed, the proportion of shared parameters between critical tasks, such as building footprint extraction and edge detection, becomes minimal. This low ratio impedes effective information sharing across subnetworks. Moreover, as the designated sparse rate diminishes, the variability in the proportion of shared parameters among tasks increases significantly, ranging up to the interval [0,1]. This results in inconsistent effects and uncontrollability when employing varying sparse rates.

Consequently, it is imperative to refine the generation approach of subnetworks by meticulously controlling the sparse rate and parameter sharing rate. Such strategic adjustments facilitate improved management of information sharing and task allocation within the multitask processing framework, thereby enhancing overall network performance.

Based on the above considerations, the randomly generated subnetwork is optimized as a constrained randomly generated subnetwork (as shown in Fig. 1). Based on the sparse rate p, the parameter sharing rate q is set to realize the control of multitask random subnetwork generation. For the two subnetworks of building footprint extraction and edge detection, the calculation formulas of sparse rate and parameter sharing rate are shown in the following equation:

$$\begin{cases} \dfrac{M_i}{M_{\text{Base}}} = p \\ \dfrac{M_i}{M_{\text{Seg}} \cap M_{\text{Edge}}} = q \end{cases} \tag{2}$$

where $M_i \in \{M_{\text{Seg}}, M_{\text{Edge}}\}$; M_{Base} is the mask of the base network, in which $M_{\text{Base}} \in \{1\}$; p is the sparse rate; and q is the parameter sharing rate.

B. Cosine Unequal-Scaled Alternating Training Strategy

In the process of building extraction, the importance of tasks, such as footprint extraction and edge detection, varies significantly. Footprint extraction classifies based on pixel analysis, while edge detection is predicated on gradient changes. Within the framework of deep learning, both tasks can be conceptualized as pixel-based classifications. Practically, edge detection can be considered a subset of footprint extraction because morphological erosion processing of footprint labels yields edge information. Thus, although edge detection is a subset of footprint extraction, it provides critical boundary information typically absent in footprint extraction tasks.

Inspired by the cosine annealing learning rate, we optimize the alternating training strategy in sparse sharing into a cosine unequal-scaled alternating training strategy. Considering the cosine unequal-scaled alternating training strategy for multitasks in the underfitting training phase, the significant performance difference on two or more tasks may slow the

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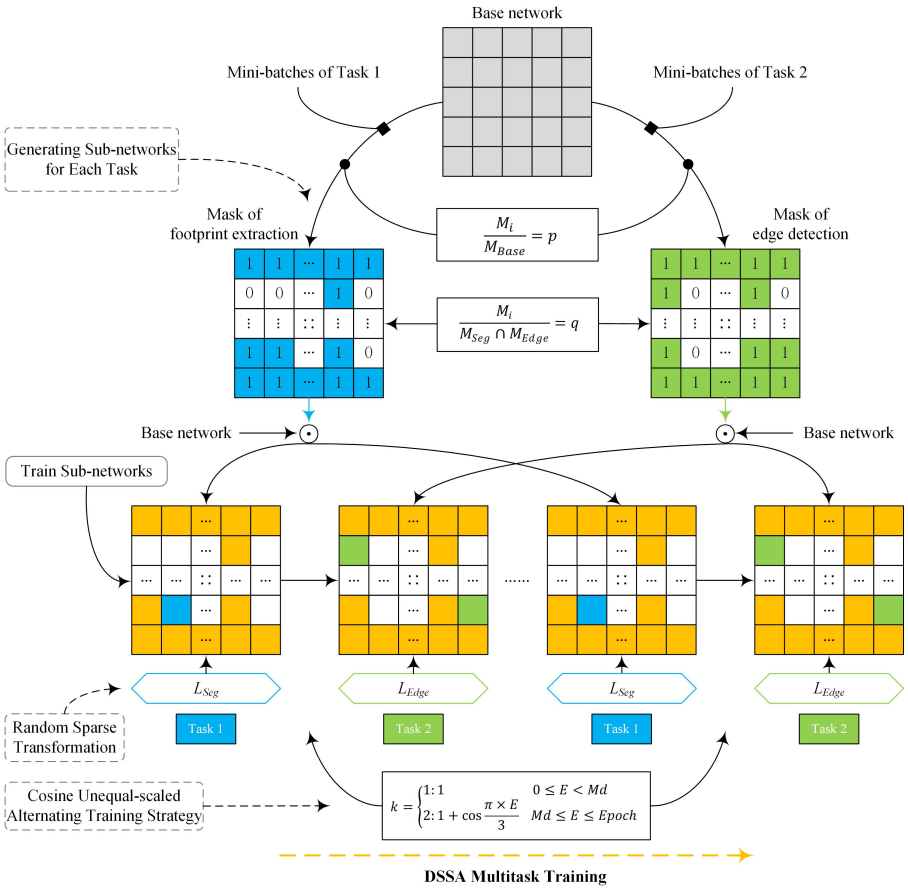

Fig. 1. Overall schematic of DSSA, orange squares represent the shared parameters; green and blue represent the respective parameters of footprint extraction and edge detection, respectively; and ⊙ denotes the element-wise multiplication.

overall network fitting time. Therefore, the equal alternating training strategy is still used in the early stage of network training, and then, the cosine unequal-scaled alternating training strategy is used after the network accuracy of multiple subnetworks has reached the specified requirements. In each training epoch, the ratio of alternating iterations between the building footprint extraction and edge detection satisfies (3). It can be seen that before the training epoch Md, the two tasks are trained uniformly alternating. After the training epoch Md, the cosine unequal-scaled alternating training strategy is used, in which the one with more iteration is considered the main task and the one with fewer iterations is considered the auxiliary task. In the case of more than two tasks, more than one task can be selected as the central task or more than one task as the auxiliary task and inherit the corresponding number of iterations

$$k = \begin{cases} 1:1 & 0 \le E < \text{Md} \\ 2:1+\cos\dfrac{\pi \times E}{3} & \text{Md} \le E \le \text{Epoch} \end{cases} \tag{3}$$

where Md is the threshold of epoch, E is denoted as E iteration epoch, and Epoch is the maximum number of iterations epoch.

Md is determined by specific accuracy evaluation Metrics in the training process and satisfies the following equation:

$$\text{Md} = \min\left(\text{Ep}\left(\text{Mt}^{\text{Seg}} \ge \text{Sc}^{\text{Seg}} \& \text{Mt}^{\text{Edge}} \ge \text{Sc}^{\text{Edge}}\right)\right) \tag{4}$$

where Mt^{Seg} denotes the score of the footprint extraction metrics, Mt^{Edge} denotes the score of the edge detection metrics, Sc^{Seg} denotes the threshold set by the footprint extraction metrics, Sc^{Edge} denotes the threshold set by the edge detection metrics, Md denotes the building footprint extraction metrics, and the edge detection metrics are greater than the minimum epoch of the corresponding score.

C. Random Sparse Network Transformation

Sparse sharing architecture can use the iterative magnitude pruning (IMP) method based on the lottery ticket hypothesis to train the sparse network, but it often requires a multitask warmup to achieve optimal results. Therefore, we employ the RST method without multitask warmup in DLTH to train the subnetwork. To implement the RST, it is necessary to utilize information from the entire network to assist the randomly selected subnetworks in performing the transformation. We refer to the widely used regularization term in DLTH to borrow the information of other parameters to implement RST.

$L3/2$ regularization, by applying a 1.5 power penalty to weights, provides a balanced approach that mitigates overfitting while maintaining model flexibility. It offers moderate sparsity, more effectively pushing some weights toward zero than $L2$, but less drastically than $L1$, thereby enhancing robustness without excessive sensitivity to outliers. This method effectively balances complexity reduction and model effectiveness, making it particularly suitable for handling

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diverse or nonuniform datasets. By incorporating an $L3/2$ regularization term into the loss function, we borrow information from other weights set to be masked, compressing this information into the targeted sparse structure. This approach facilitates the training of sparse networks. Given a randomly chosen subnetwork $F_i = F(D; M_i \odot \boldsymbol{\theta})$ with constraints from a randomly initialized multinetwork, $M_i \in \{0, 1\}$, information squeezing of subnetworks can be optimized by the following losses:

$$L_{\mathrm{BCE}} = \frac{[(1-\beta) y_i log(\hat{y}_i) + \beta(1-\boldsymbol{y_i}) \log(1-\hat{y}_i)]}{-N} \tag{5}$$

$$L_{\mathrm{Seg}} = (L_{\mathrm{BCE}})_{\boldsymbol{\theta}_{\boldsymbol{M_{Seg}=1}}} + \lambda |_{\boldsymbol{\theta}_{\boldsymbol{M_{Seg}=0}}}|^{\frac{3}{2}} \tag{6}$$

$$L_{\mathrm{Edge}} = (L_{\mathrm{BCE}})_{\boldsymbol{\theta}_{\boldsymbol{M_{Edge}=1}}} + \lambda |_{\boldsymbol{\theta}_{\boldsymbol{M_{Edge}=0}}}|^{\frac{3}{2}} \tag{7}$$

where L_{Seg} represents the loss function used for the building footprint extraction task, while L_{Edge} denotes the loss function employed for the edge detection task. β denotes the proportion of positive samples, $\boldsymbol{y_i}$ denotes the true label of the ith sample, and $\hat{y}_i$ denotes the predicted probability of the ith sample. $\boldsymbol{\theta}_{\boldsymbol{M_{Seg}=1}}$ denotes the parameter for $M_{\mathrm{Seg}} = 1$, $\boldsymbol{\theta}_{\boldsymbol{M_{Seg}=0}}$ denotes the parameter for $M_{\mathrm{Seg}} = 0$, $\boldsymbol{\theta}_{\boldsymbol{M_{Edge}=1}}$ denotes the parameter for $M_{\mathrm{Edge}} = 1$, and $\boldsymbol{\theta}_{\boldsymbol{M_{Edge}=0}}$ denotes the parameter with $M_{\mathrm{Edge}} = 0$.

Both loss functions comprise an auto weight binary cross-entropy (BCE) loss function combined with an $L3/2$ regularization term. The auto weight BCE loss function periodically trains the parameters $\boldsymbol{\theta}$ corresponding to $M_i = 1$. The $L3/2$ regularization term also incorporates the parameters $\boldsymbol{\theta}$ associated with $M_i = 0$ into the network training, but their influence is progressively suppressed by increasing the regularization. The tradeoff parameter λ is set to gradually increase from a small value during optimizing L_{seg} and L_{Edge}

$$\lambda_{a+1} = \begin{cases} \lambda_a + \eta, & \lambda_a < \lambda_b \\ \lambda_a, & \lambda_a = \lambda_b \end{cases} \tag{8}$$

where λ_a is the a update of the regularization term. η is a given small step that gradually increases the value of λ. λ_b is bound that limits the increase of λ. In this way, the size of $\boldsymbol{\theta}_{\boldsymbol{M_i=0}}$ decreases from regular to small scale as the training process progresses. As a result, the importance of the weights dynamically adjusts from the balanced (all weights are equal) to the unbalanced case $\boldsymbol{\theta}_{\boldsymbol{M_i=0}}$ becomes less important as the magnitude decreases. Information from $\boldsymbol{\theta}_{\boldsymbol{M_i=0}}$ is gradually squeezed into $\boldsymbol{\theta}_{\boldsymbol{M_i=1}}$. Similarly, random tickets are converted into winning tickets by using the information of other tickets in the whole lottery pool. After sufficient squeezing, $\boldsymbol{\theta}_{\boldsymbol{M_i=0}}$ has minimal effect on the network, and we remove them to obtain the final sparse network.

D. Comparisons With Sparse Sharing Architecture

The difference between sparse sharing architecture and DSSA is shown in Fig. 2. Based on the lottery ticket hypothesis, sparse sharing architecture mainly uses IMP. DSSA belongs to sparse sharing architecture but uses the RST based on the DLTH and optimizes the information extrusion

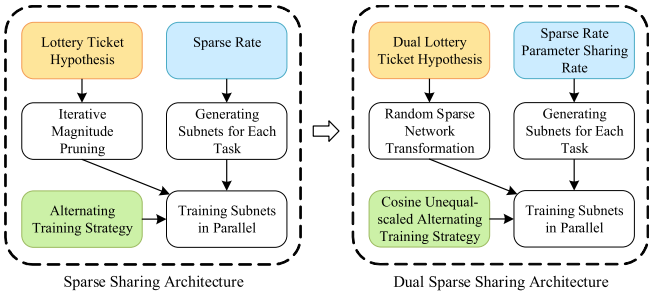

Fig. 2. Diagram of sparse sharing architecture and DSSA.

method. Sparse sharing architectures generally require multitask warmup to get better results. DSSA does not require warmup and can be trained directly to get better results. Sparse sharing does not constrain the sparse rate between multitasks in an iteration. DSSA directly constrains the parameter sharing rate between multitasks to ensure that information can be shared effectively between multitasks. After reaching a certain accuracy, the DSSA periodically strengthens and weakens the interactions between multitasks using the cosine unequal-scaled alternating training strategy.

IV. Experiments

A. Dataset

In the experiments, the proposed approach was applied to three challenging building footprint extraction datasets, i.e., the WHU aerial imagery dataset [8], the Massachusetts dataset [52], and a self-annotated building dataset. The details of these datasets are described as follows.

1) WHU Aerial Imagery Dataset: The spatial resolution of the WHU aerial imagery dataset is 0.3 m, which contains more than 187 000 buildings and covers 450 km^2 in Christchurch, New Zealand. In this dataset, each imagery has three bands, corresponding to red, green, and blue bands. There are a total of 8188 tile images, of which 4736, 2416, and 1036 tiles are used as training, test, and verification datasets, respectively. We conducted experiments on the original dataset, which provides partitions. In order to ensure that the dataset can be used on our framework, we use morphological erosion to process the original label to obtain the label of the edge detection. Samples of the cropped images are shown in Fig. 3.

2) Massachusetts Building Dataset: The Massachusetts building dataset consists of 151 aerial images from the Boston area. Each image has 1500 × 1500 pixels, a resolution of 1 m, and an area of 2.25 km^2. The data are divided into a training set of 137 images, a test set of 10, and a validation set of four images. The target map is obtained by rasterizing the building contours obtained from the OpenStreetMap project. In order to facilitate network training, we first fill the imagery to 1536 × 1536 pixels, then cut the training set according to the size of 512 × 512 pixels and the overlap rate of 50%, and remove the image with a null value ratio greater than 10% from the training dataset. The test and verification datasets are cut according to the size of 512 × 512 pixels. Thus, the training, test, and validation datasets contain 3014, 90, and 36 tiles, respectively. Edge detection labels are also generated

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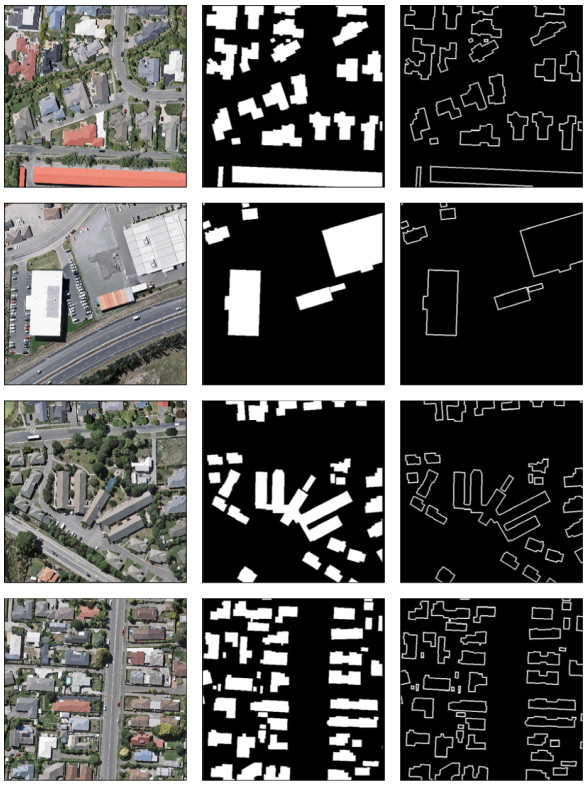

Fig. 3. Samples of the WHU aerial imagery dataset. (a) Original aerial images. (b) Building footprint labels. (c) Edge detection labels.

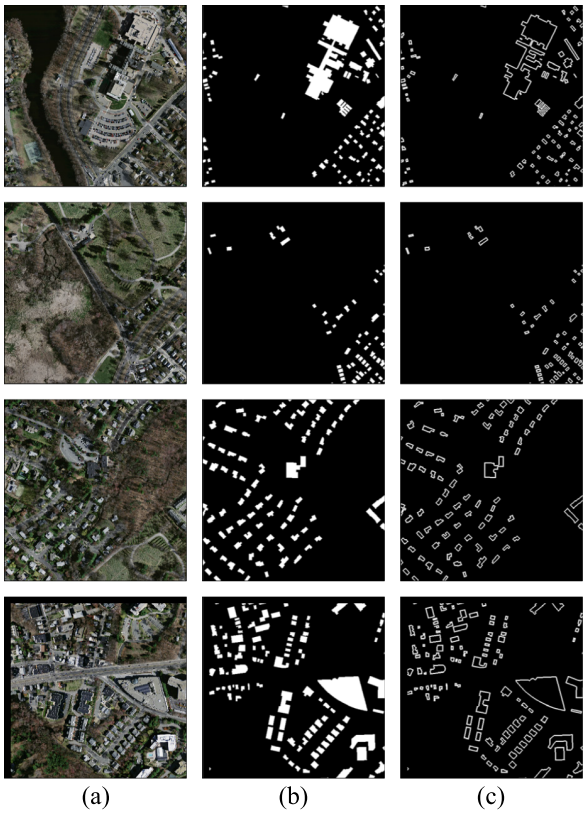

Fig. 4. Samples of the Massachusetts building dataset. (a) Original aerial images. (b) Building footprint labels. (c) Edge detection labels.

using morphological erosion. Samples of the cropped images are shown in Fig. 4.

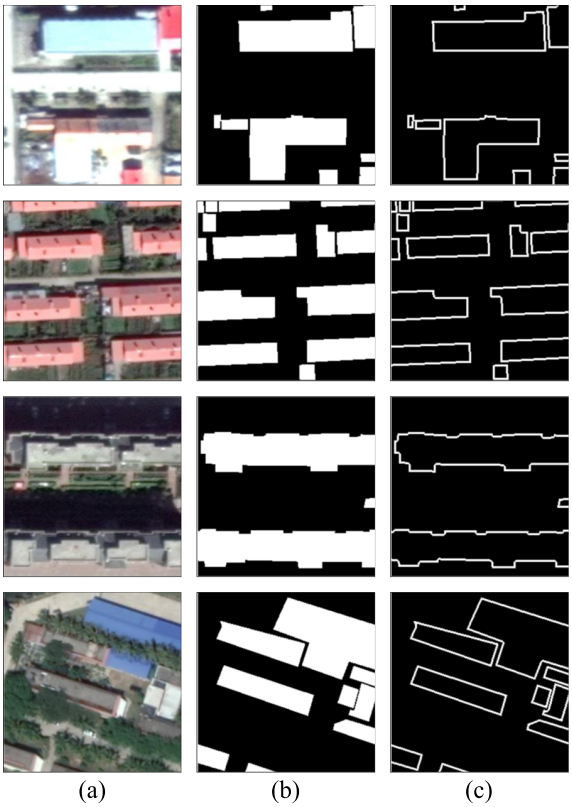

Fig. 5. Samples of the self-annotated building dataset. (a) Original aerial images. (b) Building footprint labels. (c) Edge detection labels.

3) Self-Annotated Building Dataset: The authors annotated and built this dataset independently. Remote sensing experts develop a labeling scheme and then vectorize all buildings in the target area. The remote sensing imagery of the building dataset is from Google Earth, and the spatial resolution is about 0.54 m. It covers Sanya City and Haikou City in Hainan Province, Harbin City in Heilongjiang Province, Changchun City, and Shuangya City in Jilin Province. The training data are cut into a fixed size of 256 × 256 pixels at an overlap rate of 0.5, and 19 271 sets of tiles are obtained. The verification set is cut into a fixed size of 256 × 256 pixels, and 1375 sets of tiles are obtained. Since the self-annotation dataset consists of remote sensing images taken in different regions and under different weather conditions, therefore, the dataset is challenging as well as valuable. Edge detection labels are also generated using morphological erosion. Some samples from the self-annotated building dataset are shown in Fig. 5.

B. Implementation Details

In order to verify the effectiveness of the DSSA, Hed-Unet [53] and MAP-Net [44] are used as the base network. Experiments are carried out on previously mentioned datasets to demonstrate that the proposed method is effective in multiple differentiated scenarios. The batch size is 8 for the method, and the input image size is the original size of all the datasets. The network weights were optimized by an Adam optimizer using a learning rate of 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$ hyperparameters. The hardware used for the experiments has the following specifications: an Intel[1]

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Xeon[1] Bronze 3204 CPU at 1.90 GHz, 32 GB of RAM, and one NVIDIA GeForce RTX 4090 GPU, with Windows 11 Pro for Workstations as the operating system and Python 3.7 and PyTorch 1.12.0 as the programming environments.

C. Evaluation Metrics

In order to evaluate the performance of DSSA comprehensively and quantitatively, precision, kappa, $F1$ Score, mIoU, and $\text{EEA}^{(n)}$ are used as the evaluation metrics for building footprint extraction in this study. Here, precision is the proportion of correctly predicted building pixels among all predicted building pixels. Kappa is used to measure the consistency and accuracy between the footprint results of the networks and the actual labels. $F1$ Score is a combined metric considering recall and precision, the ratio of intersection, and union of predicted building pixels and actual building pixels. mIoU is the average of IoU of building and background, and Kappa reflects the accuracy of building and background

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{10}$$

$$F1\ \text{Score} = \frac{2\times\text{Recall}\times\text{Precision}}{\text{Recall}+\text{Precision}} \tag{11}$$

$$\text{mIoU} = \frac{1}{2}\left(\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} + \frac{\text{TN}}{\text{TN}+\text{FN}+\text{FP}}\right) \tag{12}$$

$$\begin{cases} P_0 = \dfrac{\text{TP}+\text{FP}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\ P_e = \dfrac{(\text{TP}+\text{FN})(\text{TP}+\text{FP})+(\text{FN}+\text{TN})(\text{FP}+\text{TN})}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\ \text{Kappa} = \dfrac{P_0-P_e}{1-P_e} \end{cases} \tag{13}$$

where TP represents the number of correctly predicted building pixels, TN represents the number of correctly predicted background pixels, FP represents the number of background pixels incorrectly predicted to be building pixels, and FN represents the number of building pixels incorrectly predicted to be background pixels.

To better evaluate the edge effects of building footprint extraction results, pixel-based $\text{EEA}^{(n)}$ is introduced. As shown in Fig. 6, utilizing morphological image processing, the building footprint labels are eroded inward to retain n layers of pixels. The number of pixels N with the type of building after erosion is calculated. The eroded labels are intersected with the model's predicted results, and the number of pixels N_T correctly classified as building edges is calculated. Finally, N_T is divided by$N$ to obtain the $\text{EEA}^{(n)}$

$$\text{EEA}^{(n)} = \frac{N_T}{N}. \tag{14}$$

$\text{EEA}^{(n)}$'s theoretical value ranges from 0 to 1, and a larger value indicates a better network edge effect. However, in both open-source and nonopen-source datasets, some of their labels do not fit well with the actual edges of buildings, and it is often difficult to obtain high values when evaluating them.

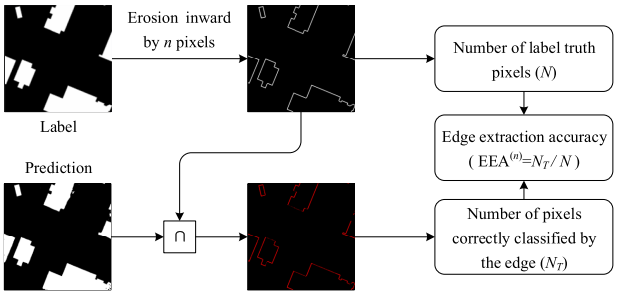

Fig. 6. Overall flow of the $\text{EEA}^{(n)}$ calculation.

V. Experimental Results and Discussion

A. Experiment Using the WHU Aerial Imagery Dataset

Fig. 7 shows the experimental results. We choose four representative images, as shown in Fig. 7(c)–(i). FCN [54], SegNet [55], U-Net [14], PSPNet [56], MAP-Net [44] and HRNet-OCR [57], Hed-Unet [53], MTL homoscedastic segmentation, reconstructed and boundary (SRB) [47], and DSSA based on Hed-Unet were used for comparative experiments. The second column in Fig. 7 is the labels of ground truth, and the following columns are the test results obtained by different networks.

The WHU aerial imagery dataset distribution is more complex, including buildings of different sizes, layouts, materials, and shapes. The prediction results based on the WHU aerial imagery dataset are shown in Fig. 7, where the columns represent the results of the network and the rows represent the samples of different buildings. Fig. 7 shows that when dealing with buildings in complex scenes, the DSSA proposed in this article can effectively inherit the feature extraction ability of the base network and can promote the base network to extract more complete building edges, and the boundary shape segmentation effect is better. It has greater advantages than other test methods. The first row shows the extraction effect of each method in the sparse area of buildings. The Hed-Unet network can effectively suppress the false detection situation. The DSSA based on the Hed-Unet network can also suppress the false detection situation, and the extracted building edges are neater. The second row shows large-scale and circular buildings and the circular buildings extracted by DSSA are rounder and more regular. The third row shows the dense area of buildings, and DSSA can completely segment the small buildings.

In this study, common footprint extraction evaluation metrics are used to validate the accuracy of DSSA on the WHU aerial imagery dataset. In addition, edge evaluation metrics are used to validate the advantages of the method in enhancing the edge morphology of buildings. The results of the quantitative evaluation are shown in Table I. The rows show the evaluation metric results for each method. The columns indicate the tested methods. The last row records the results of each metric for the methods proposed in this study. The best results are shown in bold. Specifically, DSSA (Hed-Unet) obtained the highest,Kappa, $F1$ score, and mIoU, which were 81.75%, 82.67%, and 87.15%, respectively. The $\text{EEA}^{(2)}$ of DSSA (Hed-Unet) is 0.93% higher than that of Hed-Unet. This further verifies that our method can improve the integrity of the

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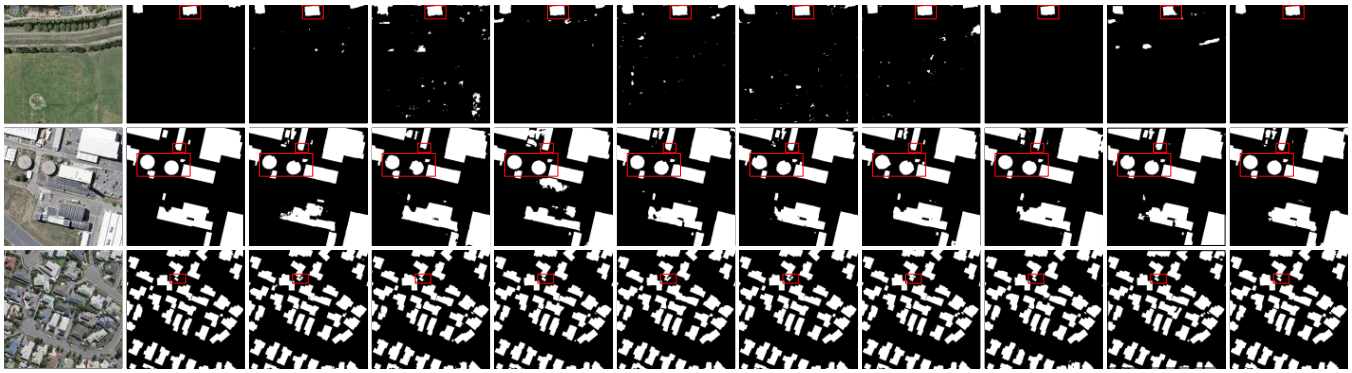

Fig. 7. Building footprint extraction results on the WHU aerial imagery dataset. (a) Images. (b) Labels. (c) FCN. (d) SegNet. (e) U-Net. (f) PSPNet. (g) MAP-Net. (h) HRNet-OCR. (i) Hed-Unet. (j) MTL homoscedastic SRB. (k) DSSA (Hed-Unet).

TABLE I
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE WHU DATASET

Strategy	Precision (%)	Kappa (%)	F1 Score (%)	mIoU (%)	$EEA^{(2)}$ (%)
FCN	77.82	79.78	80.56	86.26	**38.12**
SegNet	72.23	74.83	75.83	82.68	36.87
U-Net	71.53	74.20	75.12	82.74	36.03
PSPNet	72.71	75.47	76.38	83.09	32.35
MAP-Net	73.99	76.39	77.23	83.85	37.86
HRNet-OCR	75.07	77.23	78.04	84.55	36.92
Hed-Unet	79.36	80.20	81.23	86.06	36.43
MTL homoscedastic SRB	**81.97**	80.67	81.55	86.45	36.79
DSSA (Hed-Unet)	80.15	**81.75**	**82.67**	**87.15**	37.36

building. On this dataset, Hed-Unet's overall performance was inferior to the multitask model MTL homoscedastic SRB. However, with the integration of DSSA, DSSA (Hed-Unet) surpassed the performance of MTL homoscedastic SRB in kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, with respective improvements of 1.08%, 1.12%, 0.7%, and 0.57%. Quantitative analysis results demonstrate that the method proposed in this study has advantages in edge learning and surpasses some open-source multitask building extraction models. DSSA can optimize the boundaries of buildings and achieve precise building footprint results.

B. Experiment Using the Massachusetts Building Dataset

The Massachusetts Buildings dataset has a low resolution and is dominated by small, dense buildings, making edge segmentation of its small buildings difficult. Roofs of different materials and shading make complete segmentation of its large buildings more difficult. Compared to other mainstream single-task and multitask methods, DSSA (MAP-Net) optimizes building edges to exhibit more regular shape features. Simultaneously, DSSA (Hed-Unet) shows clearer and more complete edges on small buildings compared to Hed-Unet, and the adhesion of buildings in dense areas is also improved. In Fig. 8, the first row contains the extraction results of buildings in complex environments, the second row includes the extraction results of large buildings, and the third row indicates the results of small building extraction. Due to shadows and tree occlusions, other networks have more misdetections and omissions, which can be effectively mitigated by our method. For the problem of blurred edge segmentation of small buildings, DSSA can enhance the accuracy of small building edges from the footprint results. For large buildings, the results of DSSA extraction are more complete. Compared with other methods, this method has obvious advantages.

Quantitative evaluation results are shown in Table II, where rows represent different testing methods and columns represent different evaluation metrics. DSSA (MAP-Net) achieved the best scores in all metrics except precision. DSSA (MAP-Net) obtained the highest kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, outperforming the second-best results by 0.13%, 0.05%, 0.19%, and 2.2%, respectively. In addition quantitative results indicate that compared to MAP-Net, DSSA (MAP-Net) significantly improves edge integrity, with a 2.2% increase in $EEA^{(2)}$. Compared to Hed-Unet, DSSA (Hed-Unet) showed improvements of 5.47%, 1.77%, 1.36%, 1.37%, and 3% in precision, kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, respectively, with a notable increase in $EEA^{(2)}$. It is noteworthy that Hed-Unet's overall performance on this dataset was significantly lower than the multitask model MTL homoscedastic SRB, but DSSA (Hed-Unet) outperformed MTL homoscedastic SRB in all metrics except precision. These quantitative analyses further validate that DSSA ensures the integrity of buildings in areas of low resolution and high density of small buildings.

C. Experiment Using the Self-Annotated Building Dataset

To further test the generalization performance of DSSA regarding extracting buildings in different datasets, this study also includes experiments on the self-annotated building dataset. The distribution of the self-labeled building dataset is more complex; it covers Sanya City and Haikou City in Hainan Province, Harbin City in Heilongjiang Province, Changchun City, and Shuangya City in Jilin Province. Therefore, using this dataset to test the generalization ability of DSSA is very convincing. Due to the lack of memory in the device, the training

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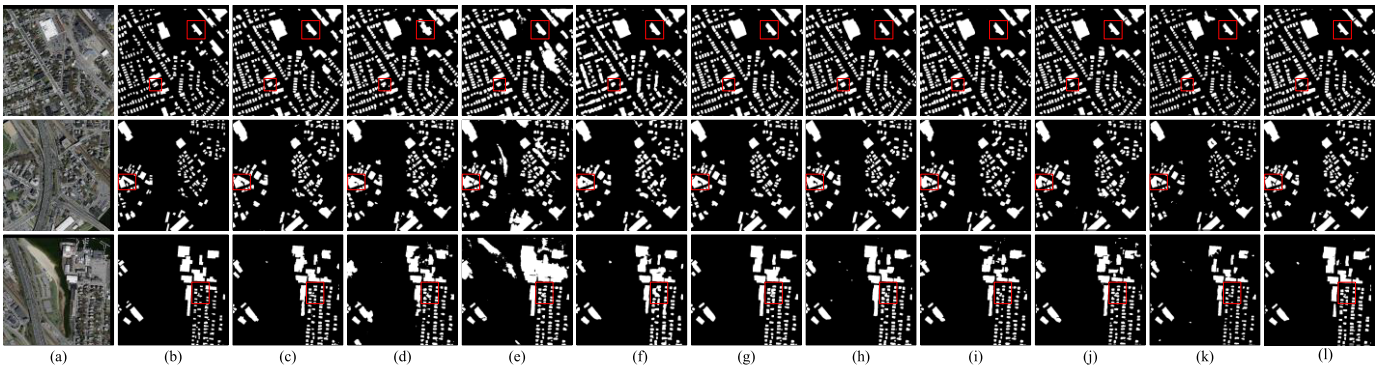

Fig. 8. Building footprint extraction results on the Massachusetts building dataset. (a) Images. (b) Labels. (c) FCN. (d) SegNet. (e) U-Net. (f) PSPNet. (g) MAP-Net. (h) HRNet-OCR. (i) Hed-Unet. (j) DSSA (Hed-Unet). (k) MTL homoscedastic SRB. (l) DSSA (MAP-Net).

TABLE II
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE MASSACHUSETTS DATASET

Strategy	Precision (%)	Kappa (%)	F1 Score (%)	mIoU (%)	$EEA^{(2)}$ (%)
FCN-8s	71.19	74.44	77.30	78.79	28.81
SegNet	67.06	71.23	74.48	76.62	26.62
U-Net	68.08	73.34	76.43	78.02	28.07
PSPNet	66.20	71.74	74.99	76.90	21.82
MAP-Net	70.72	75.52	78.27	79.55	28.98
HRNet-OCR	70.50	74.46	77.31	78.84	28.65
Hed-Unet	67.24	72.64	75.85	77.47	27.04
DSSA (Hed-Unet)	72.71	74.41	77.21	78.84	30.04
MTL homoscedastic SRB	**77.67**	73.94	76.63	78.60	30.09
DSSA (MAP-Net)	72.82	**75.65**	**78.32**	**79.74**	**31.18**

dataset is cropped into 256 × 256 pixel images according to 50% overlap, and the validation dataset is cropped into 256 × 256 pixel images according to 0% overlap. It can be seen from Fig. 9 that compared with other methods, the results of DSSA have a more complete and smoother boundary, even in complex scenes with irregular building shapes and significant differences in styles. From the results of the first row of Fig. 9, when facing low-rise buildings, DSSA can accurately segment adjacent buildings, and the segmentation result is smoother. From the results of the second row, DSSA can accurately identify the roof contour of the building when facing large buildings, effectively suppress the false detection caused by the tilt of the building, and significantly improve the accuracy of the building footprint extraction results. At the same time, other methods have many false detections. The results of the third row show that DSSA can extract the uneven boundaries of buildings and avoid false detection caused by building tilt.

In this study, a quantitative evaluation of the self-annotated building dataset was conducted, and the evaluation results are shown in Table III. DSSA achieved the highest scores in precision, kappa, $F1$ Score, and mIoU, which were 77.05%, 65.66%, 70.62%, and 73.48%, respectively. Compared with baseline, DSSA improves precision, kappa, $F1$ Score, mIoU, and $EEA^{(2)}$ by 3.51%, 2.55%, 2.21%, 1.55%, and 1.6%, respectively. Conversely, the multitask model MTL homoscedastic SRB experienced a decline in performance on this dataset, with all metrics falling below those of mainstream models. These results further show that our multitask deep learning framework can further improve the accuracy of building footprint extraction through edge detection auxiliary tasks and can achieve superior performance in identifying buildings.

In summary, even in the case of poor quality of remote sensing images and tilted buildings, DSSA can accurately identify the roof contour of buildings and suppress the interference of tilted walls of buildings. When there are concave and convex boundaries in the irregular shape of buildings, DSSA can effectively identify these boundary features and obtain more accurate boundary results than other methods. These examples prove that DSSA can cope with various complex scenarios and has strong generalization ability. At the same time, it can be proved that under DSSA, building footprint extraction can make full use of the information of edge detection and can significantly improve the recognition of building boundaries.

D. Ablation Experiment

This study includes ablation experiments on the WHU aerial imagery dataset to better understand the impact of each structure in the proposed DSSA. The output of the base network Hed-Unet has two channels, one for footprint extraction results and the other for edge detection results. In this study, two sets of ablation experiments are set up. In the first set of experiments, as shown in Table IV, the first row is the Hed-Unet using only building footprint extraction off the edge detection task, which corresponds to Fig. 10(c); the second row is the original Hed-Unet, which corresponds to Fig. 10(d); the third row is the direct conversion of the Hed-Unet to the sparse sharing parameter sharing method, which corresponds to Fig. 10(e); the fourth line is based on the third line and uses the $L3/2$ regularization term to facilitate the transformation

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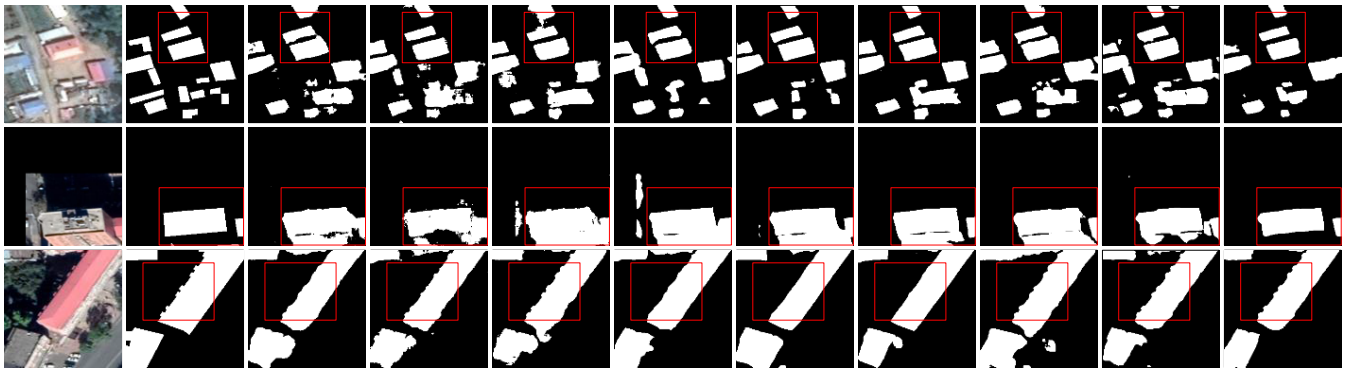

(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 9. Building footprint extraction results on the self-annotated building dataset. (a) Images. (b) Labels. (c) FCN. (d) SegNet. (e) U-Net. (f) PSPNet. (g) MAP-Net. (h) HRNet-OCR. (i) Hed-Unet. (j) MTL homoscedastic SRB. (k) DSSA (Hed-Unet).

TABLE III

COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE SELF-ANNOTATED BUILDING DATASET

Strategy	Precision (%)	Kappa (%)	F1 Score (%)	mIoU (%)	$EEA^{(2)}$ (%)
FCN-8s	68.32	64.14	69.54	72.66	17.43
SegNet	69.33	61.77	67.08	71.15	18.01
U-Net	63.80	59.06	64.48	69.77	17.59
PSPNet	66.28	60.60	65.91	70.67	14.94
MAP-Net	66.54	61.20	66.25	70.85	**18.51**
HRNet-OCR	66.04	61.02	66.23	70.89	18.20
Hed-Unet	73.54	63.11	68.41	71.93	15.60
MTL homoscedastic SRB	68.37	56.28	60.94	70.23	14.25
DSSA (Hed-Unet)	**77.05**	**65.66**	**70.62**	**73.48**	17.20

of the subnetwork, corresponding to Fig. 10(f); the fifth line introduces an additional building footprint extraction task on top of the fourth line's method, corresponding to Fig. 10(g); and the sixth line initiates the cosine unequal-scaled alternating training strategy on top of the fifth line, corresponding to Fig. 10(h).

The results of the qualitative analysis are presented in Table IV. Compared to the single-task baseline in the first row, Hed-Unet experienced a decrease in precision by 6.42%, which is because the dataset has a negative sample proportion is more, and the network tends to predict negative cases. However, improvements in kappa, $F1$ Score, and mIoU were observed at 0.99%, 0.82%, and 1.04%, respectively, indicating the effectiveness of using edge detection for assisted training. Notably, in the evaluation metric $EEA^{(2)}$, which signifies new edge integrity, Hed-Unet demonstrated a significant improvement of 8.61%, underscoring the substantial enhancement in the network's edge extraction capability through the utilization of edge information. In the third row, Hed-Unet was directly transitioned to a sparse sharing parameter sharing method. Compared to the original Hed-Unet, a decline in all performance metrics was observed, due to the difficulty in optimizing results through the direct training of randomly selected subnetworks. The fourth row builds upon the third, employing the $L3/2$ regularization term to facilitate subnetwork transformation. Thanks to the $L3/2$ regularization term, which borrows information from previously unselected parameters, there were increases of 2.25%, 6.68%, 6.26%, 4.63%, and 3.94% in precision, kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, respectively, nearly matching the performance of the original Hed-Unet. The fifth row incorporates the DSSA method with building footprint extraction set as an auxiliary subtask. This approach improved precision, kappa, $F1$ Score, and mIoU by 0.82%, 0.95%, 0.85%, and 0.65%, respectively, while a decrease of 1.16% was noted in $EEA^{(2)}$. This indicates that integrating the footprint extraction auxiliary subtask within the DSSA architecture effectively enhances the accuracy of footprint extraction, though at the expense of reduced edge completeness. The sixth row elaborates on the results following the introduction of the cosine unequal-scaled alternating training strategy based on the fifth row's approach. Subsequent to this strategy's activation, improvements in mIoU and $EEA^{(2)}$ were noted at 0.16% and 0.99%, respectively, with declines in precision, kappa, and $F1$ Score. It is important to note that compared to the second row, there was a decrease in precision by 0.18%, while kappa, $F1$ Score, mIoU, and $EEA^{(2)}$ improved by 1.04%, 0.94%, 0.79%, and 0.07%, respectively. This demonstrates that the cosine unequal-scaled alternating training strategy can effectively mitigate the adverse impacts of information transfer between auxiliary subtasks, thus enhancing overall network performance.

Fig. 10 presents the results of several ablation experiments. The first row displays extraction results from large buildings, where Fig. 10(c) shows poor extraction performance in the shadow areas where buildings meet, and Fig. 10(d) extracts uneven building edges, while the direct transition of Hed-Unet to sparse sharing architecture results in significant leakage detection issues. Although Fig. 10(e) and (g) improves on this leakage, issues still persist. In contrast, Fig. 10(h) yields the results that are more consistent with the true label and are more complete. The second row, in Fig. 10(g) and (h), demonstrates that the cosine unequal-scaled alternating training strategy can reduce oversegmentation in the main task network and effectively segment smaller buildings. The third

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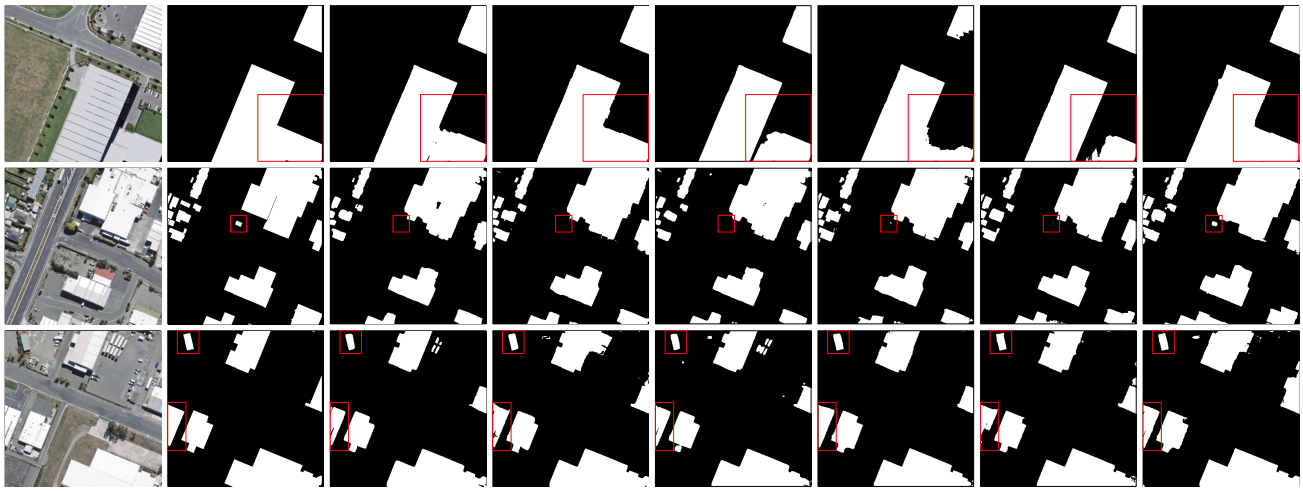
TABLE IV

INFLUENCES OF DIFFERENT MODULES IN DSSA ON THE WHU AERIAL IMAGERY DATASET UNDER THE BUILDING FOOTPRINT EXTRACTION SUBTASK

Mul	$L\frac{3}{2}$	Seg	Cos	Precision	Kappa	F1 Score	mIoU	$EEA^{(2)}$
×				**85.78**	79.21	80.41	85.02	27.82
—				79.36	80.20	81.23	86.06	36.43
√				77.64	73.68	75.15	81.41	30.41
√	√			79.89	80.36	81.41	86.04	34.35
√	√	√		80.71	**81.31**	**82.26**	86.69	35.51
√	√	√	√	79.18	81.24	82.17	**86.85**	**36.50**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 10. Ablation comparison results for building footprint extraction by DSSA, with building footprint extraction as an additional subtask. (a) Images. (b) Labels. (c) Hed-Unet (only-seg). (d) Hed-Unet. (e) Hed-Unet (Mul). (f) Hed-Unet (Mul+$L3/2$). (g) Hed-Unet (Mul+$L3/2$+Seg). (h) Hed-Unet (Mul+$L3/2$+Seg+Cos).

row, in Fig. 10(g), shows higher consistency with the true label, but Fig. 10(h) provides more complete results. Compared to Fig. 10(d), Fig. 10(e) significantly inhibits building misdetections, resulting in more accurate footprint extraction outcomes. This further reinforces the conclusions drawn from the aforementioned qualitative analysis.

In the second set of experiments, as shown in Table V, the first row is the Hed-Unet using only building footprint extraction off the edge detection task, corresponding to Fig. 11(c); the second row is the original Hed-Unet, corresponding to Fig. 11(d); the third row is the direct conversion of the Hed-Unet to the sparse sharing parameter sharing approach, corresponding to Fig. 11(e); the fourth line is based on the third line and uses the $L3/2$ regularization term to facilitate the transformation of the subnetwork, corresponding to Fig. 11(f); the fifth line introduces an additional edge detection task on top of the fourth line's method, corresponding to Fig. 11(g); and the sixth line initiates the cosine unequal-scaled alternating training strategy on top of the fifth line, corresponding to Fig. 11(h).

The results of the qualitative analysis are presented in Table V. Compared to the fourth row, which employed the $L3/2$ regularization term, the fifth row, which further incorporates an edge detection task, shows the increases of 0.4%, 0.33%, 0.43%, and 3.68% in kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, respectively, although there is a decrease in precision. Notably, after the addition of the edge detection task, there is a significant improvement in $EEA^{(2)}$, indicating that DSSA can effectively utilize information from the edge detection task to enhance the edge performance of building footprint extraction. The sixth row reflects the outcomes following the implementation of the cosine unequal-scaled alternating training strategy based on the fifth row, showing improvements of 1.62%, 0.99%, 0.93%, and 0.68% in precision, kappa, $F1$ Score, and mIoU, respectively, with a 0.67% decrease in $EEA^{(2)}$. It is important to note that compared to the original model in the second row, there are improvements across all metrics including precision, kappa, $F1$ Score, mIoU, and $EEA^{(2)}$, which have increased by 0.79%, 1.55%, 1.44%, 1.09%, and 0.93%, respectively. This demonstrates that the cosine unequal-scaled alternating training strategy effectively mitigates the negative impact of the edge detection auxiliary subtask on the footprint extraction subtask, propelling the final results to higher levels of segmentation and edge integrity. Fig. 11 showcases some results of the ablation experiments, clearly illustrating that building footprint extractions are more complete and regular in Fig. 11(g) and (h). In the first row, Fig. 11(g) and (h) is regular, but there is a small amount of missed detection internally, and Fig. 11(e) and (f) exhibits a significant amount of leakage. In the second row, Fig. 11(g) enhances the edges of the footprint results with the aid of edge information from the edge detection subtask but voids near the edges and increased false detections are present. Fig. 11(h) manages to alleviate the issues found in Fig. 11(g), with edges that are more regular and complete. In the third row, Fig. 11(h) is effective in mitigating missed detections, and the footprint results align more closely with the actual regular boundaries of the building. This also shows that the cosine unequal-scaled alternating

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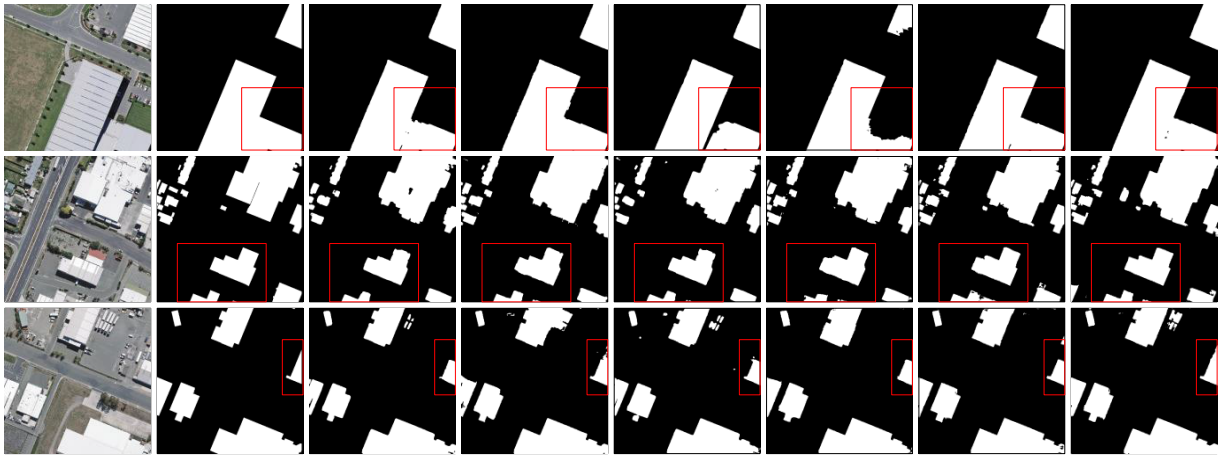
TABLE V
INFLUENCES OF DIFFERENT MODULES IN DSSA ON THE WHU AERIAL IMAGERY DATASET UNDER THE EDGE DETECTION SUBTASK

Mul	$L^{\frac{3}{2}}$	Edge	Cos	Precision	Kappa	F1 Score	mIoU	EEA(2)
×				**85.78**	79.21	80.41	85.02	27.82
—				79.36	80.20	81.23	86.06	36.43
√				77.64	73.68	75.15	81.41	30.41
√	√			79.89	80.36	81.41	86.04	34.35
√	√	√		78.53	80.76	81.74	86.47	**38.03**
√	√	√	√	80.15	**81.75**	**82.67**	**87.15**	37.36



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 11. Ablation comparison results for building footprint extraction by DSSA, with edge detection as an additional subtask. (a) Images. (b) Labels. (c) Hed-Unet (only-seg). (d) Hed-Unet. (e) Hed-Unet (Mul). (f) Hed-Unet (Mul+$L3/2$). (g) Hed-Unet (Mul+$L3/2$+Edge). (h) Hed-Unet (Mul+$L3/2$+Edge+Cos).

training strategy can solve the negative impact of information transfer between multitasks, consistent with the results of the qualitative analysis in the second set of labs.

E. Sparse Rate and Parameter Sharing Rate

To further explore the effects of sparse rate (p) and parameter sharing rate (q) on training outcomes, this research conducted two sets of comparative experiments using the Hed-Unet as the base network on the WHU aerial imagery dataset. These experiments aimed to meticulously assess the specific impacts of p and q on model performance under various settings. The first set used a fixed p value of 0.8, with q varying at 0.80, 0.85, 0.90, 0.95, and 1.00. The second set maintained a constant q of 0.9, while p was adjusted at 0.5, 0.6, 0.7, 0.8, and 0.9. All other settings were consistent with Section V-A.

The results of the first set of experiments are presented in Table VI. In summary, under the condition of a sparse rate (p) of 0.8, the parameter sharing rate (q) of 0.85 achieved the best results, showing superior performance in precision, kappa, $F1$ Score, and EEA, with respective values of 79.21%, 80.93%, 81.88%, and 38.24%. In terms of mIoU, it achieved a result of 86.60%, which was second only to a parameter sharing rate (q) of 1. The combined results at parameter sharing rates (q) of 0.80, 0.85, 0.90, and 1.00 were superior to those of the original Hed-Unet.

The results of the second set of experiments, as shown in Table VII, indicate that under a parameter sharing rate (q) of 0.9, the optimal overall results were achieved with a sharing rate (p) of 0.7, where precision, kappa, $F1$ Score, and mIoU reached 81.92%, 81.92%, 82.85%, and 87.17%, respectively. This suggests that having more parameters in subnetworks is not always beneficial. As the sparse rate (p) decreased to 0.5, there was a significant drop in accuracy, indicating that too few parameters in subnetworks can lead to performance degradation. The combined results at sparse rates (p) of 0.6, 0.7, and 0.8 were superior to those of the original Hed-Unet.

In addition, based on the sparse rate, the computational complexity of DSSA is evaluated in terms of parameters, floating-point operations per second (FLOPs), and multiply–accumulate operations (MACs). The parameters, FLOPs, and MACs of Hed-Unet are 8.13 million (M), 16.66 billion (G), and 16.77 billion (G), respectively. When the sparse rate (p) is set to 0.9, the parameters, FLOPs, and MACs of the subnetwork in DSSA (Hed-Unet) are 7.32M, 14.8G, and 14.91G, respectively, which are 90%, 88.84%, and 88.91% of the base model. This indicates that DSSA can generate subnetworks with a corresponding proportion of parameters according to sparse rate (p) while also reducing computational complexity to a certain extent.

VI. DISCUSSION

The DSSA multitask deep learning framework proposed in this article can use the rich edge information in building edge detection to make the results of building footprint extraction smoother and more complete. Compared with the classical building footprint extraction method, DSSA's building footprint extraction can pay more attention to the edge features of buildings, help the network capture more edge information, and achieve consistent results on three datasets.

Experiments on the WHU aerial imagery dataset show that DSSA (Hed-Unet) can inherit the recognition ability of the

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TABLE VI
EFFECT OF DSSA WITH p OF 0.8 DIFFERENT q'S ON THE WHU AERIAL IMAGE DATASET UNDER THE EDGE DETECTION SUBTASK

parameter sharing rate (q)	Precision (%)	Kappa (%)	F1 Score (%)	mIoU (%)	$EEA^{(2)}$ (%)
0.80	78.20	80.73	81.76	86.36	35.50
0.85	**79.21**	**80.93**	**81.88**	86.50	**38.24**
0.90	78.45	80.74	81.70	86.46	36.38
0.95	76.86	80.01	81.02	85.95	37.23
1.00	78.93	80.88	81.82	**86.60**	36.44

TABLE VII
EFFECT OF DSSA WITH q OF 0.9 DIFFERENT p'S ON THE WHU AERIAL IMAGE DATASET UNDER THE EDGE DETECTION SUBTASK

sparse rate (p)	Precision (%)	Kappa (%)	F1 Score (%)	mIoU (%)	$EEA^{(2)}$ (%)
0.50	72.73	73.30	74.30	83.11	17.91
0.60	78.90	81.11	82.06	86.58	**37.36**
0.70	**81.92**	**81.92**	**82.85**	**87.17**	35.01
0.80	78.45	80.74	81.70	86.46	36.38
0.90	78.64	80.63	81.57	86.46	35.65

baseline network Hed-Unet and can effectively avoid false detection. In the identification of circular buildings more rounded and more in line with the basic shape of the building. In the Massachusetts building dataset, due to the edge information transmitted from edge detection to building footprint extraction during the training process, DSSA (MAP-Net) can accurately segment small buildings compared to the baseline network MAP-Net, effectively alleviating the situation of multiple small buildings connected, and EEA^2 increased by 2.2%. The imagery of self-annotated building dataset comes from Google Earth. The imagery is more accessible to obtain, more representative, and more challenging. The experiments show that the extraction result of DSSA (Hed-Unet) is more complete and more regular on the edge than the baseline network Hed-Unet, and it can also reduce false detection. The ablation experiments show that the subnetwork under the constraints of sparse rate and parameter sharing rate, combined with the $L3/2$ regularization constraint loss function, can effectively learn the global information and make the network have higher generalization ability. The cosine unequal-scaled alternating training strategy can effectively alleviate the interference of one task to another and improve the performance of another task. It can effectively avoid negative transfer to achieve the excellent effect of building footprint extraction.

Although DSSA has achieved good results in building footprint extraction, there are still some limitations.

1) In the training process, a cosine unequal-scaled alternating training strategy will repeatedly switch the training data of building footprint extraction and edge detection, which takes more time and increases the time cost.
2) When DSSA is applied to specific tasks and randomly generates subnetworks, it needs to be adapted and optimized according to the network structure, which is difficult.

VII. CONCLUSION

This article proposes a sparse sharing multitask deep learning framework DSSA combined with DLTH, to solve the problem of blurred boundaries and incomplete edges in building footprint extraction from high-resolution remote sensing imagery. Our method first uses the sparse rate and parameter sharing rate to construct a subnetwork of two tasks for multitask training to ensure the sharing and transmission of information between the building footprint extraction and the edge detection. Then, following the DLTH, using the loss function with $L3/2$ regularization constraint, the random subnetwork can be transformed into the optimal network without multitask warmup. Subsequently, the cosine unequal-scaled alternating training strategy is used to periodically strengthen and weaken the transmission of edge detections to building footprint extraction information, effectively avoiding the negative impact of mutual constraints between multitasks.

The experimental results on two open-source datasets and one nonopen-source dataset show that the proposed method is superior to the baseline method. The mIoU scores on the three datasets reach 87.15%, 79.74%, and 73.48%. Compared with the baseline network, mIoU increased by 1.09%, 0.19%, and 1.55%, and the $EEA^{(2)}$ increased by 0.93%, 2.2%, and 1.6%. The results of quantification and visualization show that the DSSA method proposed in this article can effectively improve the integrity of the edge contour of the building footprint extraction results in various scenarios and is more regular.

In the future, our work will focus on the following three aspects. First, the sparse rate is adaptively adjusted during training to achieve lightweight subnetwork training. Second, the general subnetwork generation method is studied to improve the compatibility of DSSA to apply to vision transform and other structures effectively. Third, the current work only studies the promotion effect of building edge detection on footprint extraction. We plan to study further the effect of DSSA on different tasks through other tasks, such as building target detection.

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